

|                                                 |                                                           |
|-------------------------------------------------|-----------------------------------------------------------|
| Program : <b>Diploma in Printing Technology</b> |                                                           |
| Course Code : <b>3108</b>                       | Course Title: <b>Advanced Printing Machine Workshop-I</b> |
| Semester : <b>3</b>                             | Credits: <b>1.5</b>                                       |
| Course Category: <b>Program Core</b>            |                                                           |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>      | Periods per semester: <b>45</b>                           |

### Course Objectives:

- To provide knowledge in printing machine parts, principles, operating procedures, materials used .
- To demonstrate single color printing using single color sheet fed offset machine.

### Course Prerequisites:

| Topic            | Course code | Course name            | Semester |
|------------------|-------------|------------------------|----------|
| Offset principle |             | Basic printing process | 2        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO n | Description                                                                       | Duration (Hours) | Cognitive Level |
|------|-----------------------------------------------------------------------------------|------------------|-----------------|
| CO1  | To identify the functional parts of a sheet fed offset machine and material used. | 3                | Applying        |
| CO2  | To identify Tools used in the printing workshop.                                  | 3                | Applying        |
| CO3  | To identify operating procedures.                                                 | 3                | Applying        |
| CO4  | To demonstrate printing using single color offset machine.                        | 32               | Applying        |
|      | Lab Exam                                                                          | 4                |                 |

**CO-PO Mapping:**

| <b>Course Outcomes</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| CO1                    | 3          |            |            |            |            |            |            |
| CO2                    | 3          |            |            |            |            |            |            |
| CO3                    | 3          |            |            |            |            |            |            |
| CO4                    | 3          |            |            |            |            |            |            |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b> | <b>Description</b>                                                                                                                                                | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>             | <b>To identify the functional parts of a sheet fed offset machine and material used.</b>                                                                          |                         |                        |
| M1.01                  | Familiarize with infeed section, printing section and delivery section of the printing machine, and materials used.                                               | 3                       | Applying               |
| <b>CO2</b>             | <b>To identify Tools used in the printing workshop.</b>                                                                                                           |                         |                        |
| M1.01                  | Familiarize with the tools, and its uses.                                                                                                                         | 3                       | Applying               |
| <b>CO3</b>             | <b>To identify operating procedures.</b>                                                                                                                          |                         |                        |
| M1.01                  | Familiarize pre make ready procedures, and practice the procedures.                                                                                               | 1.5                     | Applying               |
| M 1.02                 | Familiarize make ready procedures, and practice the procedures.                                                                                                   | 1.5                     | Applying               |
|                        | Lab Exam I                                                                                                                                                        | 2                       |                        |
| <b>CO4</b>             | <b>To demonstrate printing using single color offset machine</b>                                                                                                  |                         |                        |
| M1.01                  | To demonstrate single color printing with sheet fed offset machine-single side with specified margin and with perfect registration.                               | 12                      | Applying               |
| M1.02                  | To demonstrate single color printing with sheet fed offset machine- double side printing with perfect alignment- work and turn method and work and tumble method. | 20                      | Applying               |
|                        | Lab Exam II                                                                                                                                                       | 2                       |                        |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                 |
|------------|----------------------------------------------------------------------------------------------------------|
| T1         | Dejdas LP and Destree TM, <b>Sheet fed offset press operating GATF</b> , USA                             |
| T2         | A S Porter, <b>Manual for lithographic press operation.</b>                                              |
| R1         | C S Misra, <b>Offset technology</b> , Anupam prakashan.                                                  |
| R2         | W.H. Bureau, <b>What the printer should know about the paper</b> , GATF.                                 |
| R3         | Michael Bernard, John Peacock., <b>Handbook of printing and production.</b>                              |
| R4         | Adams J.M, Faux,D.D and Rieber L.J, <b>Printing Technology,3rd edition</b> , Delmar Publishers, NewYork. |
| R5         | Helmut Kipphan, <b>Hand book of print media</b> , Springer Science & Business Media                      |
| R6         | <b>GATF staff Solving sheet fed offset press problems</b> , Springer Science & Business Media GATF, USA  |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                        |
|--------------|------------------------------------------------------------|
| 1            | <a href="http://www.printwiki.com">www.printwiki.com</a>   |
| 2            | <a href="http://www.slideshare.com">www.slideshare.com</a> |
| 3            | <a href="http://www.manroland.com">www.manroland.com</a>   |
| 4            | <a href="http://www.heidelberg.com">www.heidelberg.com</a> |
| 5            | <a href="http://www.printing.org">www.printing.org</a>     |